CY2001 Research Attachment 1

Picture-Hanging Puzzles

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Abstract

There is nothing wrong with second place. Your best effort is all that anyone's asking for. And if you give your best and you come in second, you come in third, you come in last, it's not about winning or losing. It's about giving it everything you've got. Now, Sam has built a monument to devilry and chaos. I deserve second place, I came in second. The only crime that's been committed here is that Oscar and Ally deserve first. We should be applauding them for getting more points. But in this sick rodeo, this bizarre *fucked up* clown festival that Sam's put together, we're here celebrating what I can only describe as the sickness at the core of America. And I'm gonna get him, I'm gonna get Sam.

Acknowledgements

Thank you to my mother for birthing me and for cutting fruit for me to snack on.

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Nomenclature

$\mathbb{N}$	The set of positive integers ($\{1, 2, \ldots\}$)
n	The number of nails in a given picture-hanging puzzle
x_i	The strand or link corresponding to the i-th nail where $1 \leqslant i \leqslant n$
x_1	The leftmost nail
x_n	The rightmost nail
x_0	The strand or link corresponding to the rope of the picture frame

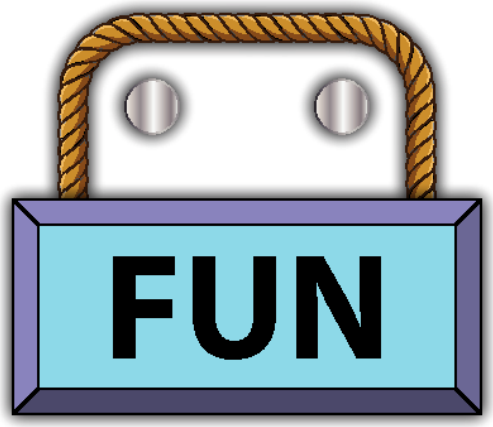


Figure 1: A normal way to hang a picture on two nails. Adapted from [4].

1 Introduction

Ordinarily, when hanging a picture frame, one would simply hang the frame by a singular nail on the wall. Obviously, removing that one nail would cause the picture and its frame to fall.

What if we were to hang the picture by two nails? Hanging the picture in the simplest way, as illustrated in Figure 1, and removing either nail would not cause the picture frame to fall. Instead, the picture frame would still hang from the remaining nail.

If we want to the picture frame to hang on both nails, but fall on the removal of either nail, how would we do so? This is the puzzle that was first set forth by A. Spivak in 1997 [1]. One such solution to the puzzle is as shown in Figure 2.

The puzzle has continued to circulate around the puzzle community, including a spot on Youtuber Tom Scott's channel in collaboration with Jade Tan-Holmes (Up and Atom) [5].

1.1 Motivation for Research

The picture-hanging puzzle asks an absurd question: how can a picture frame fall when a nail is removed despite hanging from many? The details and constraints of the puzzle will be further described in Chapter 2.

As absurd as the question may seem, there is an application for it in the Secret

Figure 2: A solution to the two-nail puzzle. Adapted from [4].

Sharing Scheme [7]. Secret Sharing in cryptography is a method for distributing a secret among a group such that no individual holds enough information to recreate the whole secret, but when a sufficient number of people share their 'secrets', they are able to recreate the whole secret [6].

Similarly, in k-out-of-n picture-hanging puzzles, the picture is expected to fall only after some requisite number of nails is pulled. Thus we can draw parallels between secret sharing and recreation and picture hanging and the fall condition.

It should be noted, however, the primary focus of this paper is on 1-out-of-n puzzles, which would be the trivial secret sharing and thus "insecure" in the real world.

1.2 Objectives

In this paper, we attempt to find a generalisation for the 1-out-of-n puzzle when the rope is allowed to wrap around itself. The construction of the wrapping is based on that proposed by Michael Paterson, featured in Demaine et al. (Figure 3)[4].

1.3 Scope of Work

This paper proposes a general result for the 1-out-of-n puzzle and proposes a potential proof for the given proposition. The proof will be based on the puzzle's connection to Borromean and Brunnian Links.

Research will focus on the construction by Michael Paterson (Figure 3) only. Other constructions where the rope wraps around itself is beyond the scope of this project.

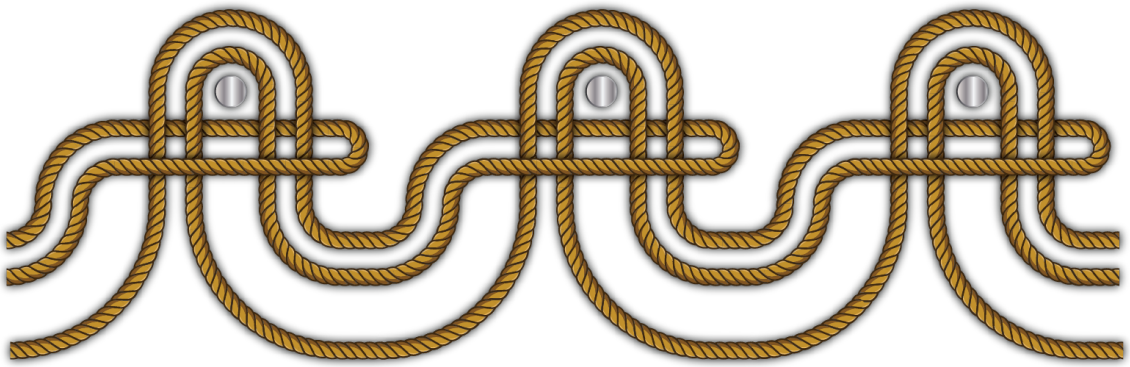


Figure 3: Construction where rope wraps around itself as proposed by Michael Paterson. Adapted from [4].

1.4 Organisation of Report

This report is organised as follows:

In Chapter 1, we have an introduction to the project and picture-hanging puzzles.

Chapter 2 summarises the findings from the literature review, including the puzzle's connection to group theory and knot theory.

Chapter 3 explains the research methodology used to conduct this research.

Chapter 4 presents the experimental model set up for this research.

Chapter 5 explains the results of this research, including the aforementioned proposition.

Chapter 6 concludes the findings of this research and recommends areas and topics for future research.

2 Literature Review

Since it's appearance in Quantum, the puzzle has continued to circulate among puzzle enthusiasts and mathematicians alike. As the puzzle circulates, others have noted certain connections between the solution to the puzzle and other mathematical concepts.

The most notable of which - for this paper at least - being that of Ed Pegg Jr. who mentioned a connection between the solution and Borromean Rings and Brunnian Links, a formalisation that will be discussed further in 2.2 [2].

First, we will discusss the formalisation using Free Group Theory.

2.1 Connection to (Free) Group Theory

This section describes a general framework to study the wrapping of the puzzle as described by Demaine et al. [4] and Ed Pegg Jr. [3]. We abstract a wrapping of a rope around n nails as a free group on n generators. Particularly, we define $2n$ symbols:

$$x_1, x_1^{-1}, x_2, x_2^{-1}, \ldots, x_n, x_n^{-1}.$$

Every x_i symbol represents a wrapping around the ith nail; x_i for a clockwise wrapping, and x_1^{-1} for an anti-clockwise one. This paper will denote the leftmost nail as x_1 and the rightmost nail as x_n. Now our wrappings can be represented by some sequence of these $2n$ symbols.

As an example, the wrapping in Figure 2 can be expressed as $x_1x_2x_1^{-1}x_2^{-1}$. This is further illustrated by Figure 4: first we wrap around x_1 clockwise, then around x_2 clockwise, then anti-clockwise around x_1^{-1}, and finally anti-clockwise around x_2^{-1}.

When removing a nail, for example the kth nail, we remove all instances of x_k and x_k^{-1} in a given sequence. With this representation, it is clear to see why this wrapping is a solution to the 2 nail puzzle. For example, removing the leftmost nail (x_1) leaves just $x_2x_2^{-1}$ and removing the rightmost (x_2) nail leaves just $x_1x_1^{-1}$. The remaining sequence is equivalent to wrapping clockwise then immediately unwrapping the rope with an anti-clockwise turn. In general, x_i and x_i^{-1} cancel out, thus any instances of $x_ix_i^{-1}$ and $x_i^{-1}x_i$ can be dropped (The free group specifies that these cancellations are all the simplifications that can be made).

Thus, the original weaving $x_1x_2x_1^{-1}x_2^{-1}$ is non-trivially linked with both nails since nothing simplifies; but removing either nail simplifies the sequence resulting in a trivial wrapping where nothing is linked (i.e. the picture falls).

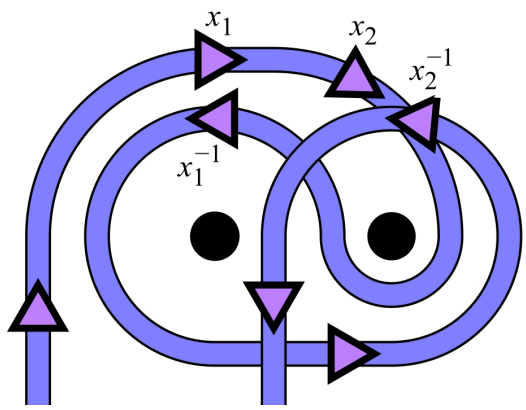


Figure 4: Understanding the notation for Figure 2. Adapted from [4].

In group theory, the expression $x_1x_2x_1^{-1}x_2^{-1}$ is called the commutator of x_1 and x_2 and is written $[x_1, x_2]$.

As defined in [4], generally, a picture hanging on n nails to be a sequence of symbols or *word* in the free group on n generators. We refer to the *length* of the word is the number of symbols in the word, this would also approximate the required length of rope to complete the wrapping that corresponds to the given word. We also define a special identity work (element) 1 that represents the picture falling. Removing the ith nail removes all instances of x_i and x_i^{-1} from the given word. Through simplification after said removal, we can determine if the picture falls.

2.2 Connection to Borromean and Brunnian Links

Werner Schwärzler observed that the solution to the two nail puzzle could be generalised to the Borromean rings. Hence, puzzles with more than two nails could be generalised further to more generalised forms of the Borromean rings. This section describes these connections.

The Borromean Rings (5) consists of three loops (rings) that by itself is inseparable - also known as non-trivially linked - but removing any one ring cause the other two to be separable [2]. These rings get its namesake from the Italian noble family Borromeo who used the rings as its crest.

Like the rings, in the picture-haning puzzle, the three loops are linked, but removing any one of them unlinks all of them. As described by Demaine et al., by stretching two of the loops to infinity and straigthening it out, we may view the loops as nails [4]. Alternatively, taking each nail and extending it out of the nail head and looping back into the plane the puzzle is nailed to, we may view the nails

Figure 5: The Borromean Rings.

as loops. Either way, we may convert the two-nail puzzle to the Borromean rings and vice versa.

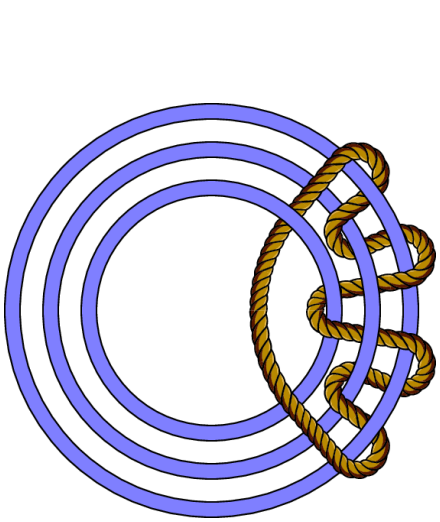
While there are two generalisations of the Borromean Rings, for this problem we will focus on the generalisation to Brunnian Links

A Brunnian link is a collection of n loops that are linked such that the removal of any one loop unlinks the rest. This is the property we want for the generalisation of our n-nail puzzle. The transformation for Brunnian 4 link to a 1-out-of-3 puzzle is shown in Figure 6

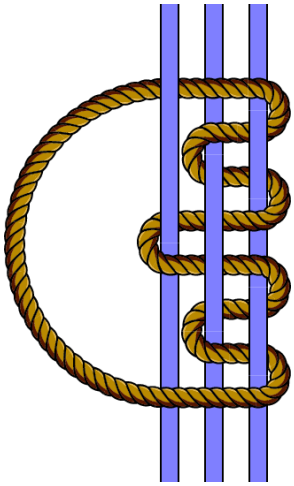
It should be noted, that this construction of the Brunnian Link fails for k-out-of-n puzzles ($k \in \mathbb{N}$). However, this paper only focuses on 1-out-of-n puzzles. Thus, the Brunnian link formalisation is adequate for this our construction where the rope wraps around itself.

2.3 Research Gaps

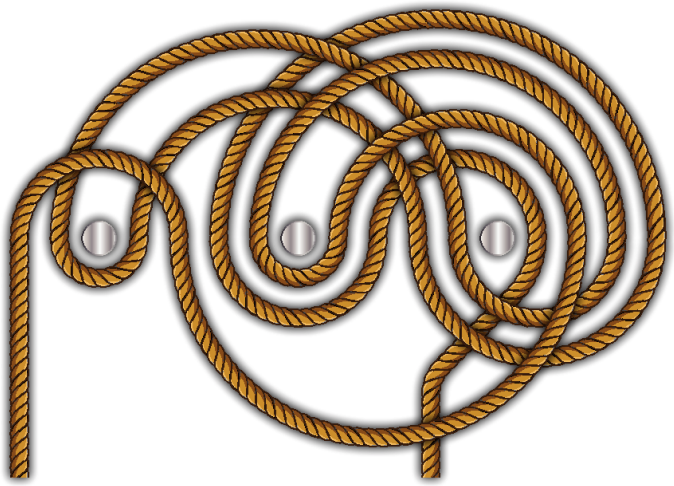
Current research gaps primarily surround optimality of typical constructions were the rope does not wrap around itself. As previously mentioned,



(a) Brunnian 4-link.



(b) Stretching all but one loop into lines.



(c) Picture-hanging equivalent from over-under pattern of (b).

Figure 6: Transforming a Brunnian n-link into a 1-out-of-n picture-hanging puzzle, for $n = 4$. Adapted from [4]

3 Research Methodology

As is the case with pure mathematical research, this research was conducted analytically.

Firstly, an extensive literature review on the topic of picture-hanging puzzles was conducted to gain pre-requisite knowledge in the field of interest. This review also served to build an understanding of particularly interesting or impactful open problems that are deserving of greater attention. Having obtained deeper intimacy with the topic, we then apply the theorems and results obtained within the papers reviewed to develop further theories in the field of picture-hanging puzzles. We also draw upon the ideas used in proofs to guide our own thoughts.

Having minimal background on the field of knot and group theory, research also entailed reading textbooks to gain relevant knowledge on these areas of study.

3.1 Scope of Investigation

As mentioned in Chapter 1, we will focus on the construction proposed by Michael Paterson (Figure 3) [4]. Our construction adapts Paterson's construction for finite values of n. Using our own construction, we check if it satisfies the 1-out-of-n puzzle requirements. If the puzzle is solved with our construction, we prove why it is so. Otherwise, we make adjustments to our construction to enable the removal of any one nail to fell the picture.

We conduct these checks and adjustments for small values of n ($n = 2, 3, 4, 5, 6$). From there, we draw patterns to form a generalisation for n nails.

4 Model Setup

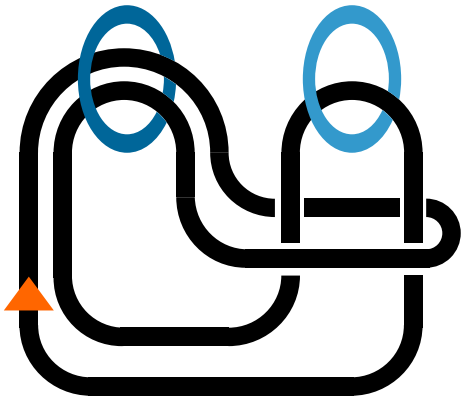
Our construction adapts Paterson's finite values of n. We begin with the construction for $n = 2$.

4.1 Original and Transformed Constructions for $n = 2$

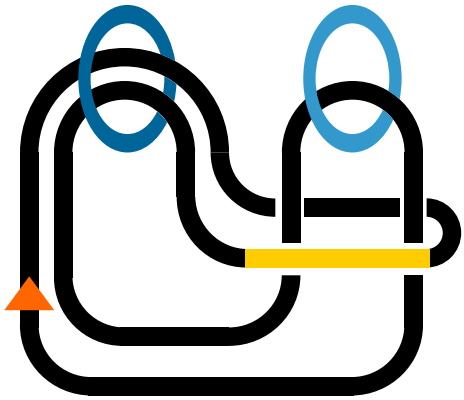
Our two-nail construction is as shown in Figure 7. First, we have the original construction (Figure 7a). We then transform our construction by taking the front section of the loop below x_2 and pulling it onto the nail head. It should be noted, that since the loops represent nails, while topologically impossible, in reality, we are in fact able to pull that section of the rope "into" the loop. This move yields the transformed construction (Figure 7d).

4.2 Original and Transformed Constructions for $n = 3$

4.3 Original and Transformed Constructions for $n \in \mathbb{N}$



(a) Original Construction for $n = 2$.

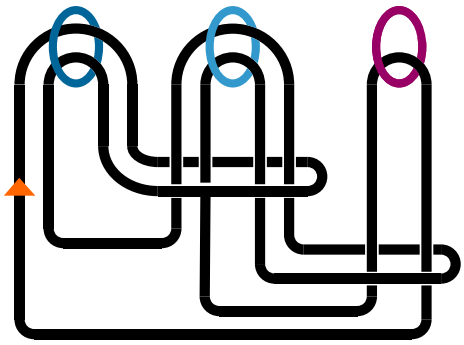


(b) Original Construction for $n = 2$. Rope section to be moved is in yellow.

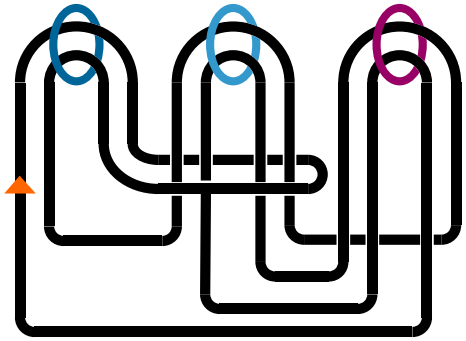
(c) Transformed Construction for $n = 2$. Moved section in yellow.

(d) Transformed Construction for $n = 2$.

Figure 7: Original and Transformed Constructions for the two-nail puzzle ($n = 2$).

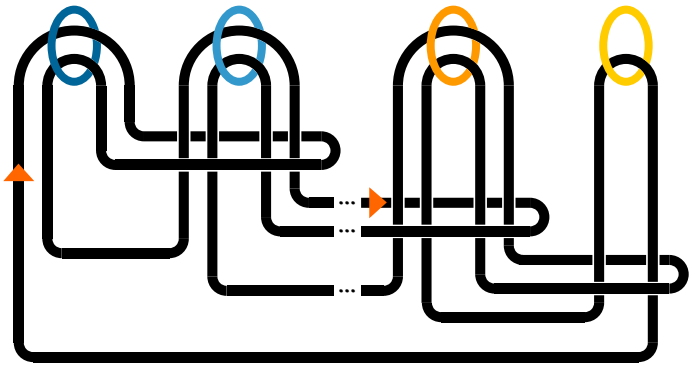


(a) Original Construction for $n = 3$

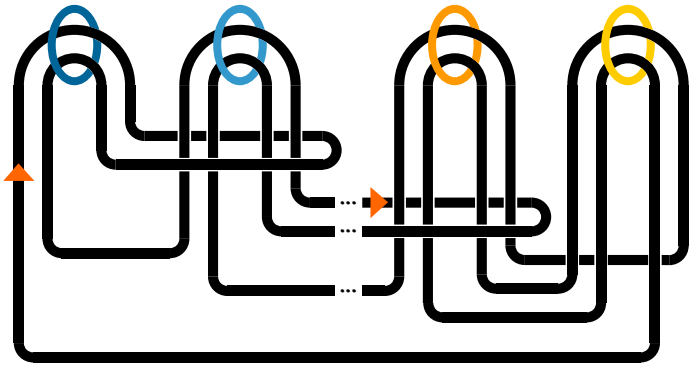


(b) Transformed Construction for $n = 3$

Figure 8: Original and Transformed Constructions for three-nail puzzle ($n = 3$).



(a) Original Construction for $n \in \mathbb{N}$



(b) Transformed Construction for $n \in \mathbb{N}$

Figure 9: Original and Transformed Constructions for n-nail puzzle ($n \in \mathbb{N}$)

5 Results

5.1 Proposition 1:

5.2 Proposition 2:

6 Conclusions and Recommendations

6.1 Conclusions

6.2 Recommendations for Future Work

References

[1] A. Spivak, "Brainteasers b 201: Strange painting," *Quantum: The Magazine of Math and Science*, p. 13, May 1997, Brainteaser problem, May–June.

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[7] R. Hu, "Picture Hanging Puzzles and Secret Sharing Scheme,"

Appendix A - Python Code for Conway Polynomial Computation

```python
from sage.all import *
from sage.plot import *
from braids import braids

for name, word in braids.items():
    print(name, ":")
    # counts number of strands, based on word
    k = int(max(abs(i) for i in word) + 1)

    B = BraidGroup(k)
    b = B(word)
    print("Number of strands:" ,b.components_in_closure())
    B(word).plot(orientation = "top-bottom").save(f"{name}_braid.png")

    L = Link(b)
    print("Number of components",L.number_of_components())
    L.plot().save(f"{name}_link.png")

    print("Conway polynomial:", L.conway_polynomial())
```

Appendix B - Dictionary of Braid Words

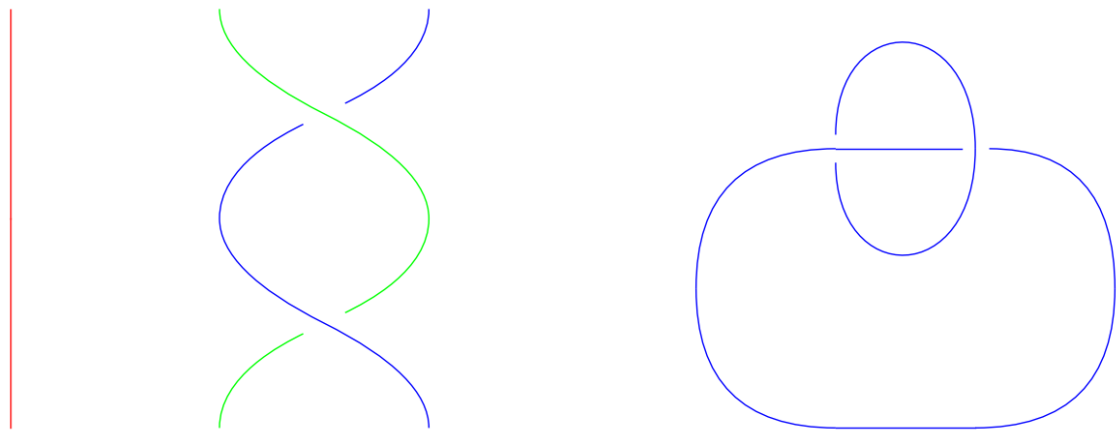
```
braids = {
    "2 original": [1, 1, -1, -1, 2, 2],
    "2 transformed": [1, 1, -2, -2, -1, -1, 2, 2],
    "3 original": [1, 1, -1, -1, 2, 2, -2, -2, 3, 3],
    "3 transformed": [1, 1, -1, -1, 2, 2, -3, -3, -2, -2, 3, 3],
    "4 original": [1, 1, -1, -1, 2, 2, -2, -2, 3, 3, -3, -3, 4, 4],
    "4 transformed": [1, 1, -1, -1, 2, 2, -2, -2, 3, 3, -4, -4, -3, -3, 4, 4],
    "5 original": [1, 1, -1, -1,
                   2, 2, -2, -2,
                   3, 3, -3, -3,
                   4, 4, -4, -4,
                   5, 5],
    "5 transformed": [1, 1, -1, -1,
                      2, 2, -2, -2,
                      3, 3, -3, -3,
                      4, 4, -5, -5, -4, -4, 5, 5],
    "6 original": [1, 1, -1, -1,
                   2, 2, -2, -2,
                   3, 3, -3, -3,
                   4, 4, -4, -4,
                   5, 5, -5, -5,
                   6, 6],
    "6 transformed": [1, 1, -1, -1,
                      2, 2, -2, -2,
                      3, 3, -3, -3,
                      4, 4, -4, -4,
                      5, 5, -6, -6, -5, -5, 6, 6],
}
```

Appendix C - Text output of Python Code in Appendix A

```
2 original :
Number of strands: 3
Number of components 2
Conway polynomial: t
2 transformed :
Number of strands: 3
Number of components 3
Conway polynomial: t^4
3 original :
Number of strands: 4
Number of components 2
Conway polynomial: t
3 transformed :
Number of strands: 4
Number of components 3
Conway polynomial: t^4
4 original :
Number of strands: 5
Number of components 2
Conway polynomial: t
4 transformed :
Number of strands: 5
Number of components 3
Conway polynomial: t^4
5 original :
Number of strands: 6
Number of components 2
Conway polynomial: t
5 transformed :
Number of strands: 6
Number of components 3
Conway polynomial: t^4
6 original :
```

```
Number of strands: 7
Number of components 2
Conway polynomial: t
6 transformed :
Number of strands: 7
Number of components 3
Conway polynomial: t^4
```

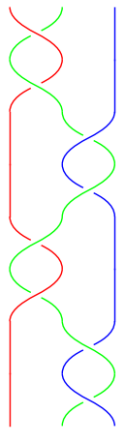
Appendix D - Figures from Python Code in Appendix A



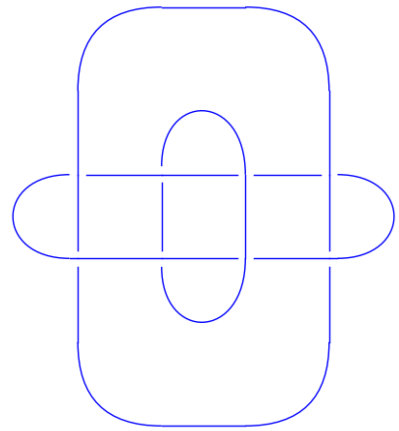
(a) Braid Diagram for $n = 2$

(b) Link Diagram for $n = 2$

Figure 10: Braid and Link Diagram for original two-nail puzzle ($n = 2$).

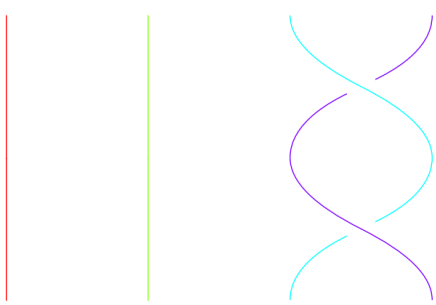


(a) Braid Diagram for $n = 2$

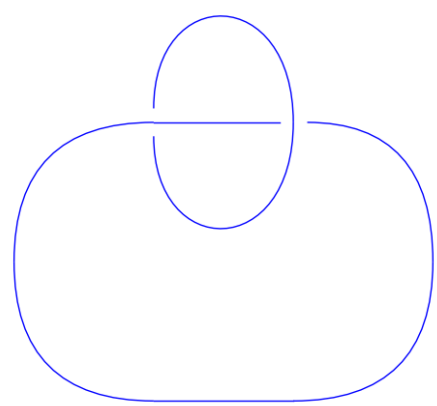


(b) Link Diagram for $n = 2$

Figure 11: Braid and Link Diagram for transformed two-nail puzzle ($n = 2$).

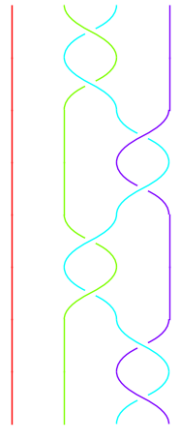


(a) Braid Diagram for $n = 3$

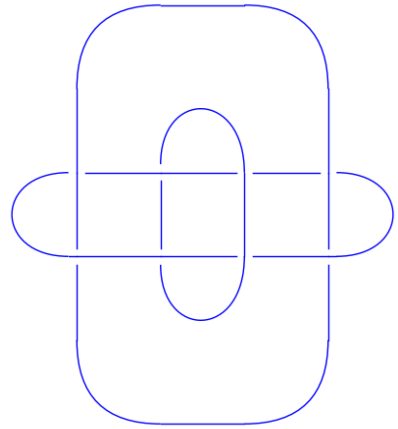


(b) Link Diagram for $n = 3$

Figure 12: Braid and Link Diagram for original three-nail puzzle ($n = 3$).

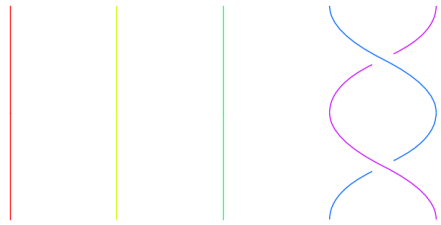


(a) Braid Diagram for $n = 3$

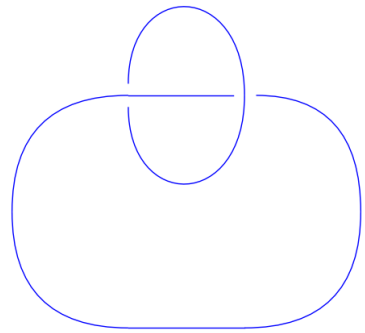


(b) Link Diagram for $n = 3$

Figure 13: Braid and Link Diagram for transformed three-nail puzzle ($n = 3$).

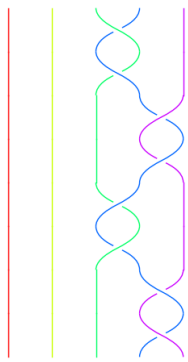


(a) Braid Diagram for $n = 4$

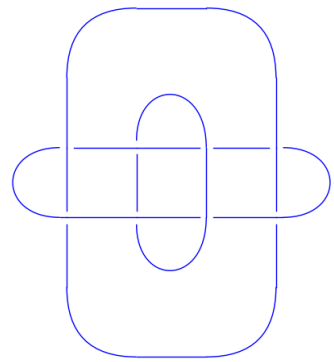


(b) Link Diagram for $n = 4$

Figure 14: Braid and Link Diagram for original four-nail puzzle ($n = 4$).

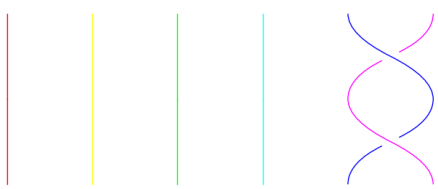


(a) Braid Diagram for $n = 4$

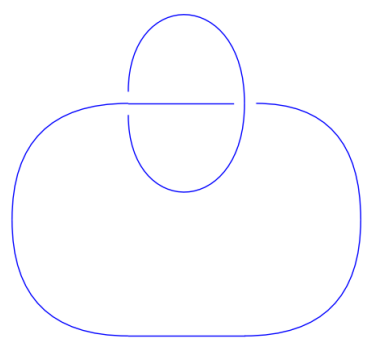


(b) Link Diagram for $n = 4$

Figure 15: Braid and Link Diagram for transformed four-nail puzzle ($n = 4$).

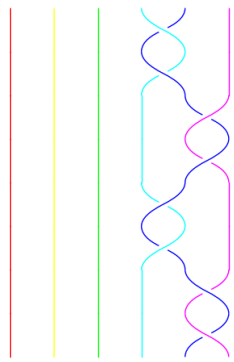


(a) Braid Diagram for $n = 5$

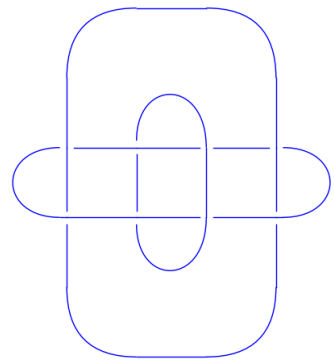


(b) Link Diagram for $n = 5$

Figure 16: Braid and Link Diagram for original five-nail puzzle ($n = 5$).

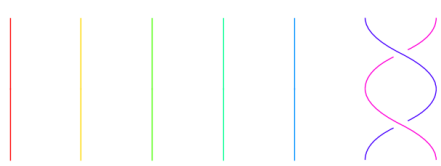


(a) Braid Diagram for $n = 5$

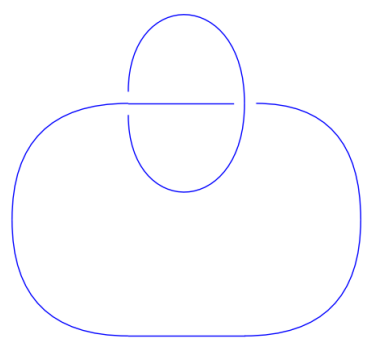


(b) Link Diagram for $n = 5$

Figure 17: Braid and Link Diagram for transformed five-nail puzzle ($n = 5$).

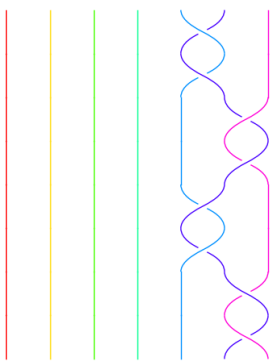


(a) Braid Diagram for $n = 6$

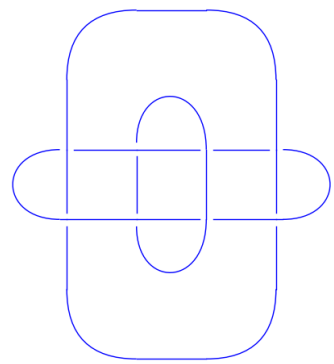


(b) Link Diagram for $n = 6$

Figure 18: Braid and Link Diagram for original six-nail puzzle ($n = 6$).



(a) Braid Diagram for $n = 6$



(b) Link Diagram for $n = 6$

Figure 19: Braid and Link Diagram for transformed six-nail puzzle ($n = 6$).